

# Seasonal Sales Analysis

**Objective:**

Analyze the difference in product sales across different seasons to determine which product categories perform best at different times of the year.

**Tools Used:**

- Python (Data Analysis and Visualization)
- Data Source: "BrewCrew Café Dataset.xlsx"

**Methodology:**

- We extracted a Season column from the Date field, categorizing transactions into Winter, Spring, Summer, and Autumn.
- We filtered out all refunded transactions (Return = 1) to ensure accuracy and only analyzed net profit.
- We grouped the cleaned data by Season and Category and calculated the total profit per group.
- A bar chart was created to visually represent seasonal performance across different product categories.

 **Key Insights (Net Profit Only):**

| Season              | Winter                                                                                       | Spring                                                                                       | Summer                                                                                       | Autumn                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Top-Selling Product |  Pastries |  Espresso |  Pastries |  Espresso |

- Pastries dominate in Winter and Summer, indicating a strong preference in cold and hot months.
- Espresso performs best in Spring and Autumn, making it an ideal focus for mid-year campaigns.
- Cold Brew is consistent but never the top performer in any season.

 **Recommendations:**

- Focus marketing efforts on Pastries during Winter and Summer.
- Promote Espresso heavily in Spring and Autumn.
- Consider bundling Cold Brew with seasonal promotions to increase visibility.